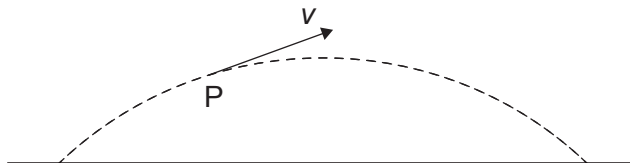


PAPER 1 / QUESTION 31

3 November 2022 / TZ0 / HL Paper 1 / original question 5 / 8822-6501

31. The diagram shows the trajectory of a projectile and the velocity v of the projectile at point P in its trajectory. P is located before the projectile reaches the peak altitude. Air resistance acts on the projectile. The acceleration of the projectile at P is a .



What are the magnitudes of the horizontal component and the vertical component of the acceleration of the projectile at P?

	Horizontal component of a	Vertical component of a
A.	zero	greater than 9.8 m s^{-2}
B.	non-zero	greater than 9.8 m s^{-2}
C.	zero	9.8 m s^{-2}
D.	non-zero	9.8 m s^{-2}